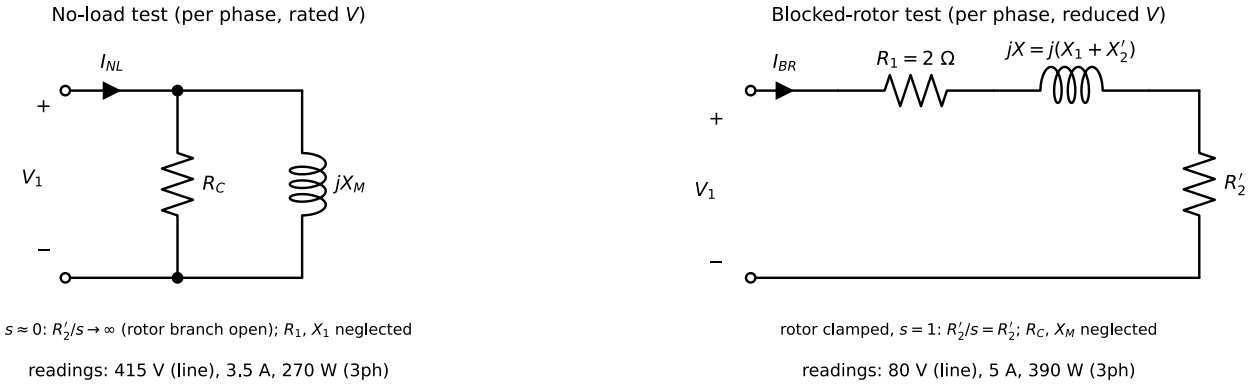
# Machine tests, maximum torque, generator mode

Practice questions · Day 11 · Tue 3 Nov · pack M8

## 5. Practice

### From your course files

**Jun24 Q3b (past exam, 4 marks).** Induction Machine Parameters. The following tests are performed on a 415V, 50Hz three-phase induction motor which is star-connected.
Stator resistance measurement: 2 ohms per phase
No-load test: input voltage 415V, input current 3.5A, input power 270W
Blocked-rotor test: input voltage 80V, input current 5A, input power 390W
Find the following simplified equivalent circuit series parameters: $R_1$, $R_2'$ and $X$.




Jun24 Q3b test circuits: no-load 415 V, 3.5 A, 270 W; blocked rotor 80 V, 5 A, 390 W; R1 = 2 ohm

**Jun23 Q3b (past exam, 5 marks).** Induction Machine Torque vs Speed. The equivalent-circuit parameters of a wound-rotor induction machine are listed below.
stator resistance $R_1 = 0.4\ \Omega$, rotor resistance $R_2' = 0.5\ \Omega$
stator leakage reactance $X_1 = 1.3\ \Omega$, rotor leakage reactance $X_2' = 1.5\ \Omega$
magnetising reactance $X_M = 95\ \Omega$, core loss resistance $R_C = 270\ \Omega$

i. Calculate the external resistance, $R_{EXT}$, required such that maximum torque occurs at standstill. [2 marks]
ii. On the axes below, draw a typical torque vs. speed characteristic curves for an induction machine – assume that this curve represents the above induction machine with zero

external resistance. In addition, draw the torque curves for external resistance equal to 0.5 and 1pu, where 1pu represents the $R_{EXT}$ calculated in part i) above. [3 marks]




Jun23 Q3b(ii) answer axes (from the exam paper): torque (labelled T on the paper) against speed n

**PQ 7.11 (practice questions).** A 2.2 kW three-phase induction machine has the following ratings: 2.2 kW, 50 Hz, 1420 rpm, 415 V, 4.9 A, 0.80 efficiency, 0.78 power-factor.

a. Calculate the synchronous speed in rpm
b. Calculate the rated output torque in Nm
c. Calculate the rated input power to the machine: i) Using the voltage, current and power-factor ii) Using the output and efficiency
d. Calculate the ideal efficiency at rated speed. Why is this different to the actual efficiency?

**PQ 7.12 (practice questions).** The following tests are performed on a 415 V, 1455rpm, 50 Hz three-phase induction motor which is star-connected.
Stator resistance measurement: 3 ohms per phase

No-load test: input voltage 415 V, input current 3.0 A, input power 205 W
Blocked-rotor test: input voltage 65 V, input current 4.8 A, input power 350 W

a. Find the simplified equivalent circuit parameters.
b. Draw the simplified equivalent circuit showing the calculated parameter values.
c. The motor is running at rated speed. Calculate the magnitude of the input current, the power-factor, the input power, output torque and output power, and hence the efficiency of the motor.
d. Calculate the rated copper losses, iron loss and total loss. Using the output power calculated in part (c), hence calculate the efficiency of the motor.

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**PQ 7.13 (practice questions).** The following tests are performed on a 415V, 1410rpm, 50Hz three-phase induction motor which is star-connected.
Stator resistance measurement: 1 ohm per phase
No-load test: input voltage 415V, input current 5.4A, input power 300W
Blocked-rotor test: input voltage 90V, input current 8A, input power 270W
Find the simplified equivalent circuit parameters.

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**PQ 7.15 (practice questions).** A three-phase, 460 V, 60 Hz, 1760 rpm, wound-rotor induction motor has the following equivalent circuit parameters
$R_1 = 0.25$ ohm : $R_2' = 0.2$ ohm : $X_1 = 1.2$ ohm : $X_2' = 1.1$ ohm : $X_M = 35$ ohm
The motor is connected to a 3ph, 460V, 60Hz supply and is star-connected. Use the simplified equivalent circuit. Initially the external rotor resistance is set to zero.

a. Calculate the speed (in rpm) at which maximum torque occurs.
b. Using the torque equation, calculate the output torque at the following speeds (i) synchronous, (ii) rated speed, (iii) speed corresponding to maximum torque, (iv) locked rotor (0 rpm)
c. Using the calculated torques in part (b), sketch out the torque versus speed characteristic.
d. The total effective rotor resistance $R_2'$ is adjusted by adding external rotor resistance. What is the total effective rotor resistance required to give maximum torque at standstill? What is the starting torque under this condition?

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**Jun24 Q3c (past exam, 10 marks).** Induction Machine Calculation. A 415 V, 50 Hz, 960 rpm, three-phase, star-connected induction machine has the following equivalent circuit parameters (referred to the stator winding where appropriate):
stator resistance $R_1 = 2.5$ ohms, rotor resistance $R_2' = 3$ ohms
stator leakage reactance $X_1 = 6$ ohms, rotor leakage reactance $X_2' = 6$ ohms
The magnetising reactance and core loss resistance can be neglected.

i. The above induction machine is run as a generator at 1025 rpm. Find the synchronous speed, the value of slip $s$ and hence the effective rotor resistance $R_2'/s$. [3 marks]
ii. Draw a single-phase simplified equivalent circuit for the machine when operating at 1025 rpm, showing the parameter values as impedances. Assuming that the machine is operating at rated voltage, find the magnitude and phase of the input phase current. Hence find the input power. You should assume that the phase angle of the phase voltage is zero. [5 marks]
iii. Using the results from part (ii) find the corresponding airgap power $P_{AG}$ using the fact that in this case $I_2' = I_1$, and hence find the output torque $\tau$. [2 marks]

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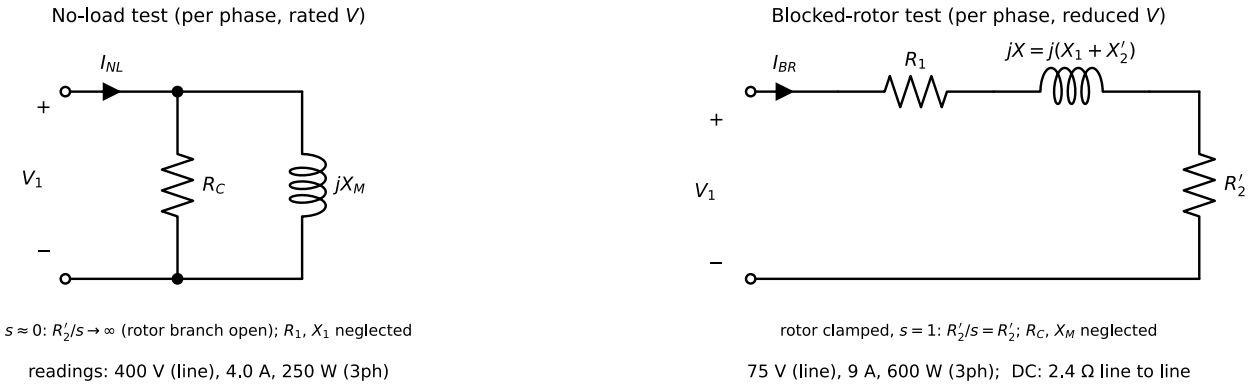
**PQ 7.14 (practice questions, if time: parts (d)–(f)).** A 415 V, 50 Hz, 2790 rpm, 3 phase, star-connected induction motor has the following equivalent circuit parameters (all referred to the stator winding): stator leakage reactance $X_1 = 0.3$ ohm, stator resistance $R_1 = 0.2$ ohms, rotor leakage reactance $X_2' = 0.3$ ohm, rotor resistance $R_2' = 0.16$ ohm and magnetising reactance $X_M = 12$ ohm. You can ignore the core-loss resistance. Use the simplified equivalent circuit for the following calculations. The motor is operated at rated voltage.

a. A locked-rotor test (slip = 1) is performed at rated voltage. Ignoring the magnetising reactance for this part only, calculate the input current magnitude, input power and output torque
b. At rated speed find the input current magnitude, input power factor and input power (do not neglect the magnetizing reactance!).
c. At rated speed find the rated output torque and output power and hence the efficiency.
d. What are the starting torque and starting current as a fraction of the rated torque and rated current respectively?
e. Determine the maximum torque available and the speed at which it occurs. What is the maximum torque as a fraction of rated torque?

f. Calculate the magnitude of the input current at no-load. What is this as a fraction of rated current?
g. Using the calculated torque at starting, maximum torque and rated torque, and your knowledge of the general shape of the induction motor torque versus speed characteristics, draw a rough sketch showing the output torque as a fraction of rated torque as a function of speed in rpm.

## New questions

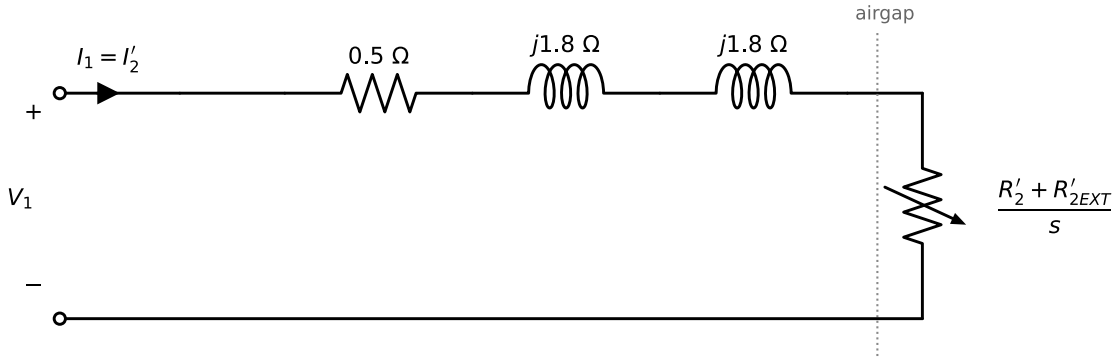
**New Q1 (easy) [5 marks].** A 400 V, 50 Hz, star-connected induction motor is tested. DC test: 2.4 Ω measured between two lines. No load: 400 V, 4.0 A, 250 W. Blocked rotor: 75 V, 9 A, 600 W. Find $R_1$, $R_2'$, $X_1$, $X_2'$, $R_C$ and $X_M$.




New Q1 test circuits: no-load 400 V, 4.0 A, 250 W; blocked rotor 75 V, 9 A, 600 W

**New Q2 (easy–medium) [6 marks].** A 415 V, 50 Hz, 6-pole, star-connected wound-rotor induction motor has $R_1 = 0.5\ \Omega$, $R_2' = 0.6\ \Omega$ and $X_1 = X_2' = 1.8\ \Omega$ (simplified circuit).

a. Find the speed at which maximum torque occurs, and the maximum torque. [3]
b. Find the starting torque and starting current (neglect the magnetising branch). [1]
c. Find the external rotor resistance (referred) that gives maximum torque at standstill, and the new starting torque and current. [2]




415 V (line), star, 6-pole; $R_2' = 0.6$ Ω; $R_{2EXT}' = 0$ except in (c)

New Q2 circuit: 0.5 ohm, j1.8 ohm, j1.8 ohm and (R2′ + R2EXT′)/s

**New Q3 (medium) [5 marks].** The New Q2 motor (no external resistance) is fed from an inverter with constant V/f.

a. What line voltage is needed at 25 Hz? [1]
b. Find the speed at maximum torque and the maximum torque at 25 Hz. Compare with 50 Hz and explain the difference. [3]
c. At 50 Hz the motor drives a 60 Nm load at 976.8 rpm. At 25 Hz it drives the same load at 475.9 rpm. What does this show about the slip speed? [1]

**New Q4 (medium) [6 marks].** The New Q2 motor (no external resistance) is driven by a wind turbine at 1040 rpm, connected to the 415 V, 50 Hz grid. Neglect the magnetising branch.

a. Motor or generator? Find $s$ and $R_2'/s$. [2]
b. Find the phase current (magnitude and angle), the real power and the reactive power at the terminals. [2]
c. Find the airgap power, the torque and the mechanical power the turbine must supply. [2]

**New Q5 (hard, chained) [10 marks].** Use the New Q1 parameters. The motor is 4-pole and rated at 1440 rpm. Use the simplified equivalent circuit at 400 V.

a. At rated speed find $I_1$ (magnitude and phase), the power factor, the input power, the output torque, the output power and the efficiency. [5]
b. Find the maximum torque, the speed at which it occurs, and $\tau_{max}/\tau_R$. Find also $\tau_{start}/\tau_R$. [3]
c. The shaft is now driven at 1530 rpm. Find the torque and say what it means. [1]
d. Find the no-load current as a fraction of the rated current. Is this typical? [1]

**New Q6 (short answer, words only) [6 marks].**

a. Carefully explain why an induction motor draws a significant current at no load, and why its power factor is low there. [3]
b. A squirrel-cage rotor is made with zero resistance. How would this change the machine's behaviour? [3]

## 6. Quick recall check

1. In the blocked-rotor test, which branch is neglected and why? Which quantity do you need from another test?
2. Write $s_{\tau max}$ in terms of the circuit parameters. What happens to $\tau_{max}$ if $R_2'$ doubles?
3. For a wound rotor, what is $R_{2EXT}'$ for maximum torque at standstill?
4. A 4-pole, 50 Hz machine runs at 1560 rpm. Give the sign of $s$, $R_2'/s$, $\tau$, $P_{IN}$ and $Q_{IN}$.
5. Under V/f control at half frequency, which three quantities in the torque formula change, and by what factor?